For the following problems, evaluate the limit using the squeeze theorem. Use a calculator to graph the functions $f(x)$, $g(x)$, and $h(x)$ when possible.

126. **[T]** True or False? If $2x-1 \le g(x) \le x^2-2x+3$, then $\lim_{x\to 2} g(x) = 0$.

127. **[T]** $\lim_{\theta\to 0} \theta^2 \cos\left(\frac{1}{\theta}\right)$

128. $\lim_{x\to 0} f(x)$, where

$$f(x) = \begin{cases} 0, & x \text{ rational} \\ x^2, & x \text{ irrational} \end{cases}$$

129. **[T]** In physics, the magnitude of an electric field generated by a point charge at a distance r in vacuum is governed by Coulomb's law: $E(r) = \frac{q}{4\pi\varepsilon_0 r^2}$, where E represents the magnitude of the electric field, q is the charge of the particle, r is the distance between the particle and where the strength of the field is measured, and $\frac{1}{4\pi\varepsilon_0}$ is Coulomb's constant: $8.988 \times 10^9\ \text{N}\cdot\text{m}^2/\text{C}^2$.

a. Use a graphing calculator to graph $E(r)$ given that the charge of the particle is $q = 10^{-10}$.

b. Evaluate $\lim_{r\to 0^+} E(r)$. What is the physical meaning of this quantity? Is it physically relevant? Why are you evaluating from the right?

130. **[T]** The density of an object is given by its mass divided by its volume: $\rho = m/V$.

a. Use a calculator to plot the volume as a function of density $(V = m/\rho)$, assuming you are examining something of mass 8 kg $(m = 8)$.

b. Evaluate $\lim_{\rho\to 0^+} V(\rho)$ and explain the physical meaning.

2.4 Continuity

Learning Objectives

2.4.1 Explain the three conditions for continuity at a point.
2.4.2 Describe three kinds of discontinuities.
2.4.3 Define continuity on an interval.
2.4.4 State the theorem for limits of composite functions.
2.4.5 Provide an example of the intermediate value theorem.

Many functions have the property that their graphs can be traced with a pencil without lifting the pencil from the page. Such functions are called *continuous*. Other functions have points at which a break in the graph occurs, but satisfy this property over intervals contained in their domains. They are continuous on these intervals and are said to have a *discontinuity at a point* where a break occurs.

We begin our investigation of continuity by exploring what it means for a function to have *continuity at a point*. Intuitively, a function is continuous at a particular point if there is no break in its graph at that point.

Continuity at a Point

Before we look at a formal definition of what it means for a function to be continuous at a point, let's consider various functions that fail to meet our intuitive notion of what it means to be continuous at a point. We then create a list of conditions that prevent such failures.

Our first function of interest is shown in Figure 2.32. We see that the graph of $f(x)$ has a hole at a. In fact, $f(a)$ is undefined. At the very least, for $f(x)$ to be continuous at a, we need the following condition:

i. $f(a)$ is defined.

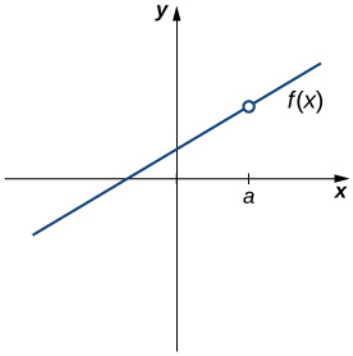


Figure 2.32 The function $f(x)$ is not continuous at a because $f(a)$ is undefined.

However, as we see in Figure 2.33, this condition alone is insufficient to guarantee continuity at the point a. Although $f(a)$ is defined, the function has a gap at a. In this example, the gap exists because $\lim_{x \to a} f(x)$ does not exist. We must add another condition for continuity at a—namely,

ii. $\lim_{x \to a} f(x)$ exists.

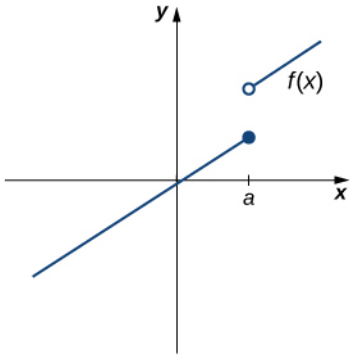


Figure 2.33 The function $f(x)$ is not continuous at a because $\lim_{x \to a} f(x)$ does not exist.

However, as we see in Figure 2.34, these two conditions by themselves do not guarantee continuity at a point. The function in this figure satisfies both of our first two conditions, but is still not continuous at a. We must add a third condition to our list:

iii. $\lim_{x \to a} f(x) = f(a)$.

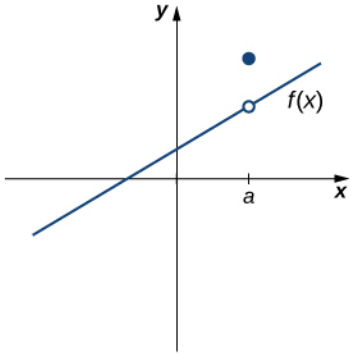


Figure 2.34 The function $f(x)$ is not continuous at a because $\lim_{x \to a} f(x) \neq f(a)$.

Now we put our list of conditions together and form a definition of continuity at a point.

Definition

A function $f(x)$ is **continuous at a point** *a* if and only if the following three conditions are satisfied:

i. $f(a)$ is defined
ii. $\lim_{x \to a} f(x)$ exists
iii. $\lim_{x \to a} f(x) = f(a)$

A function is **discontinuous at a point** *a* if it fails to be continuous at *a*.

The following procedure can be used to analyze the continuity of a function at a point using this definition.

Problem-Solving Strategy

Determining Continuity at a Point

1. Check to see if $f(a)$ is defined. If $f(a)$ is undefined, we need go no further. The function is not continuous at *a*. If $f(a)$ is defined, continue to step 2.
2. Compute $\lim_{x \to a} f(x)$. In some cases, we may need to do this by first computing $\lim_{x \to a^-} f(x)$ and $\lim_{x \to a^+} f(x)$. If $\lim_{x \to a} f(x)$ does not exist (that is, it is not a real number), then the function is not continuous at *a* and the problem is solved. If $\lim_{x \to a} f(x)$ exists, then continue to step 3.
3. Compare $f(a)$ and $\lim_{x \to a} f(x)$. If $\lim_{x \to a} f(x) \neq f(a)$, then the function is not continuous at *a*. If $\lim_{x \to a} f(x) = f(a)$, then the function is continuous at *a*.

The next three examples demonstrate how to apply this definition to determine whether a function is continuous at a given point. These examples illustrate situations in which each of the conditions for continuity in the definition succeed or fail.

EXAMPLE 2.26

Determining Continuity at a Point, Condition 1

Using the definition, determine whether the function $f(x) = (x^2 - 4)/(x - 2)$ is continuous at $x = 2$. Justify the conclusion.

Solution

Let's begin by trying to calculate $f(2)$. We can see that $f(2) = 0/0$, which is undefined. Therefore, $f(x) = \frac{x^2 - 4}{x - 2}$ is discontinuous at 2 because $f(2)$ is undefined. The graph of $f(x)$ is shown in Figure 2.35.

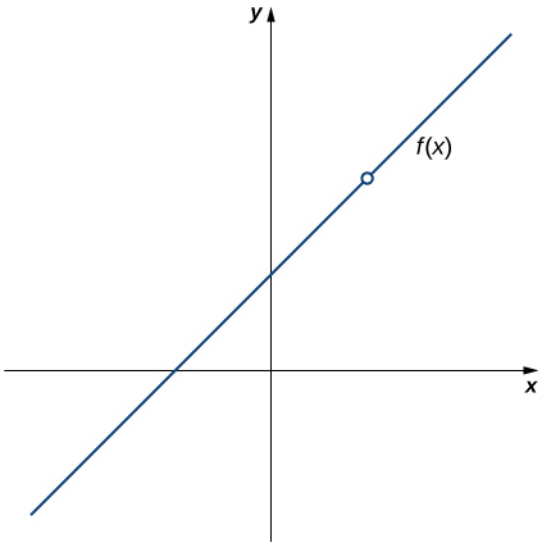


Figure 2.35 The function $f(x)$ is discontinuous at 2 because $f(2)$ is undefined.

EXAMPLE 2.27

Determining Continuity at a Point, Condition 2

Using the definition, determine whether the function $f(x) = \begin{cases} -x^2 + 4 & \text{if } x \leq 3 \\ 4x - 8 & \text{if } x > 3 \end{cases}$ is continuous at $x = 3$. Justify the conclusion.

Solution

Let's begin by trying to calculate $f(3)$.

$$f(3) = -(3^2) + 4 = -5.$$

Thus, $f(3)$ is defined. Next, we calculate $\lim_{x \to 3} f(x)$. To do this, we must compute $\lim_{x \to 3^-} f(x)$ and $\lim_{x \to 3^+} f(x)$:

$$\lim_{x \to 3^-} f(x) = -(3^2) + 4 = -5$$

and

$$\lim_{x \to 3^+} f(x) = 4(3) - 8 = 4.$$

Therefore, $\lim_{x \to 3} f(x)$ does not exist. Thus, $f(x)$ is not continuous at 3. The graph of $f(x)$ is shown in Figure 2.36.

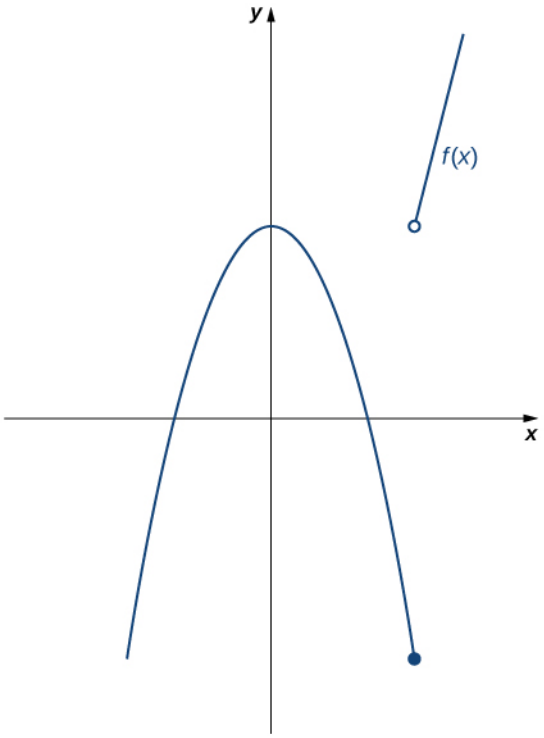


Figure 2.36 The function $f(x)$ is not continuous at 3 because $\lim_{x\to 3} f(x)$ does not exist.

EXAMPLE 2.28

Determining Continuity at a Point, Condition 3

Using the definition, determine whether the function $f(x) = \begin{cases} \frac{\sin x}{x} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$ is continuous at $x = 0$.

Solution

First, observe that

$$f(0) = 1.$$

Next,

$$\lim_{x\to 0} f(x) = \lim_{x\to 0} \frac{\sin x}{x} = 1.$$

Last, compare $f(0)$ and $\lim_{x\to 0} f(x)$. We see that

$$f(0) = 1 = \lim_{x\to 0} f(x).$$

Since all three of the conditions in the definition of continuity are satisfied, $f(x)$ is continuous at $x = 0$.

2.21 Using the definition, determine whether the function $f(x) = \begin{cases} 2x+1 & \text{if } x < 1 \\ 2 & \text{if } x = 1 \\ -x+4 & \text{if } x > 1 \end{cases}$ is continuous at $x = 1$. If the function is not continuous at 1, indicate the condition for continuity at a point that fails to hold.

By applying the definition of continuity and previously established theorems concerning the evaluation of limits, we can state the following theorem.

Theorem 2.8

Continuity of Polynomials and Rational Functions
Polynomials and rational functions are continuous at every point in their domains.

Proof

Previously, we showed that if $p(x)$ and $q(x)$ are polynomials, $\lim_{x \to a} p(x) = p(a)$ for every polynomial $p(x)$ and $\lim_{x \to a} \frac{p(x)}{q(x)} = \frac{p(a)}{q(a)}$ as long as $q(a) \neq 0$. Therefore, polynomials and rational functions are continuous on their domains.

□

We now apply Continuity of Polynomials and Rational Functions to determine the points at which a given rational function is continuous.

EXAMPLE 2.29

Continuity of a Rational Function
For what values of x is $f(x) = \frac{x+1}{x-5}$ continuous?

Solution

The rational function $f(x) = \frac{x+1}{x-5}$ is continuous for every value of x except $x = 5$.

2.22 For what values of x is $f(x) = 3x^4 - 4x^2$ continuous?

Types of Discontinuities

As we have seen in Example 2.26 and Example 2.27, discontinuities take on several different appearances. We classify the types of discontinuities we have seen thus far as removable discontinuities, infinite discontinuities, or jump discontinuities. Intuitively, a **removable discontinuity** is a discontinuity for which there is a hole in the graph, a **jump discontinuity** is a noninfinite discontinuity for which the sections of the function do not meet up, and an **infinite discontinuity** is a discontinuity located at a vertical asymptote. Figure 2.37 illustrates the differences in these types of discontinuities. Although these terms provide a handy way of describing three common types of discontinuities, keep in mind that not all discontinuities fit neatly into these categories.

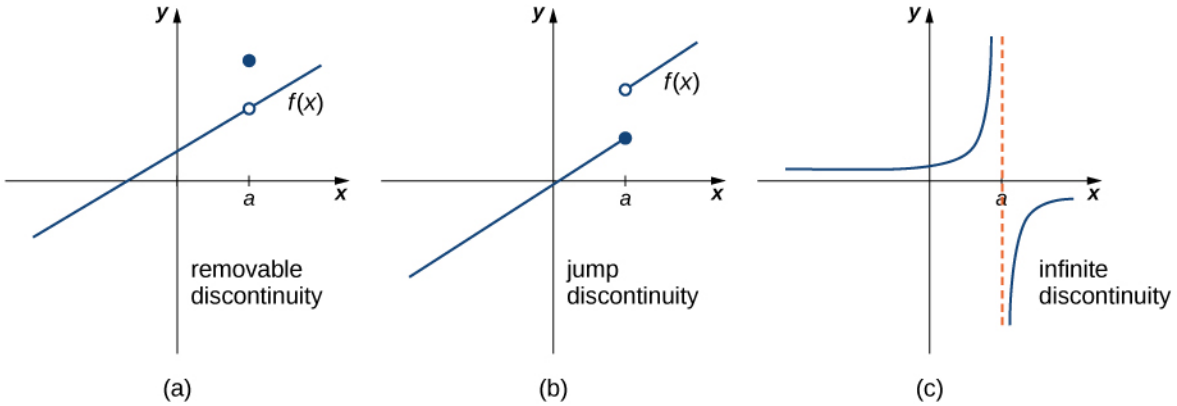


Figure 2.37 Discontinuities are classified as (a) removable, (b) jump, or (c) infinite.

These three discontinuities are formally defined as follows:

Definition

If $f(x)$ is discontinuous at a, then

1. f has a **removable discontinuity** at a if $\lim_{x\to a} f(x)$ exists. (Note: When we state that $\lim_{x\to a} f(x)$ exists, we mean that $\lim_{x\to a} f(x) = L$, where L is a real number.)
2. f has a **jump discontinuity** at a if $\lim_{x\to a^-} f(x)$ and $\lim_{x\to a^+} f(x)$ both exist, but $\lim_{x\to a^-} f(x) \neq \lim_{x\to a^+} f(x)$. (Note: When we state that $\lim_{x\to a^-} f(x)$ and $\lim_{x\to a^+} f(x)$ both exist, we mean that both are real-valued and that neither take on the values $\pm\infty$.)
3. f has an **infinite discontinuity** at a if $\lim_{x\to a^-} f(x) = \pm\infty$ and/or $\lim_{x\to a^+} f(x) = \pm\infty$.

EXAMPLE 2.30

Classifying a Discontinuity

In Example 2.26, we showed that $f(x) = \frac{x^2-4}{x-2}$ is discontinuous at $x = 2$. Classify this discontinuity as removable, jump, or infinite.

Solution

To classify the discontinuity at 2 we must evaluate $\lim_{x\to 2} f(x)$:

$$\begin{aligned}\lim_{x\to 2} f(x) &= \lim_{x\to 2} \frac{x^2-4}{x-2} \\ &= \lim_{x\to 2} \frac{(x-2)(x+2)}{x-2} \\ &= \lim_{x\to 2} (x+2) \\ &= 4.\end{aligned}$$

Since f is discontinuous at 2 and $\lim_{x\to 2} f(x)$ exists, f has a removable discontinuity at $x = 2$.

EXAMPLE 2.31

Classifying a Discontinuity

In Example 2.27, we showed that $f(x) = \begin{cases} -x^2+4 & \text{if } x \leq 3 \\ 4x-8 & \text{if } x > 3 \end{cases}$ is discontinuous at $x = 3$. Classify this discontinuity as removable, jump, or infinite.

Solution

Earlier, we showed that f is discontinuous at 3 because $\lim_{x\to 3} f(x)$ does not exist. However, since $\lim_{x\to 3^-} f(x) = -5$ and $\lim_{x\to 3^+} f(x) = 4$ both exist, we conclude that the function has a jump discontinuity at 3.

EXAMPLE 2.32

Classifying a Discontinuity

Determine whether $f(x) = \frac{x+2}{x+1}$ is continuous at −1. If the function is discontinuous at −1, classify the discontinuity as removable, jump, or infinite.

Solution

The function value $f(-1)$ is undefined. Therefore, the function is not continuous at −1. To determine the type of discontinuity, we must determine the limit at −1. We see that $\lim_{x\to -1^-} \frac{x+2}{x+1} = -\infty$ and $\lim_{x\to -1^+} \frac{x+2}{x+1} = +\infty$. Therefore, the function has an infinite discontinuity at −1.

✓ 2.23 For $f(x) = \begin{cases} x^2 & \text{if } x \neq 1 \\ 3 & \text{if } x = 1 \end{cases}$, decide whether f is continuous at 1. If f is not continuous at 1, classify the discontinuity as removable, jump, or infinite.

Continuity over an Interval

Now that we have explored the concept of continuity at a point, we extend that idea to **continuity over an interval**. As we develop this idea for different types of intervals, it may be useful to keep in mind the intuitive idea that a function is continuous over an interval if we can use a pencil to trace the function between any two points in the interval without lifting the pencil from the paper. In preparation for defining continuity on an interval, we begin by looking at the definition of what it means for a function to be continuous from the right at a point and continuous from the left at a point.

Continuity from the Right and from the Left

A function $f(x)$ is said to be **continuous from the right** at a if $\lim\limits_{x \to a^+} f(x) = f(a)$.

A function $f(x)$ is said to be **continuous from the left** at a if $\lim\limits_{x \to a^-} f(x) = f(a)$.

A function is continuous over an open interval if it is continuous at every point in the interval. A function $f(x)$ is continuous over a closed interval of the form $[a, b]$ if it is continuous at every point in (a, b) and is continuous from the right at a and is continuous from the left at b. Analogously, a function $f(x)$ is continuous over an interval of the form $(a, b]$ if it is continuous over (a, b) and is continuous from the left at b. Continuity over other types of intervals are defined in a similar fashion.

Requiring that $\lim\limits_{x \to a^+} f(x) = f(a)$ and $\lim\limits_{x \to b^-} f(x) = f(b)$ ensures that we can trace the graph of the function from the point $(a, f(a))$ to the point $(b, f(b))$ without lifting the pencil. If, for example, $\lim\limits_{x \to a^+} f(x) \neq f(a)$, we would need to lift our pencil to jump from $f(a)$ to the graph of the rest of the function over $(a, b]$.

EXAMPLE 2.33

Continuity on an Interval

State the interval(s) over which the function $f(x) = \frac{x-1}{x^2+2x}$ is continuous.

Solution

Since $f(x) = \frac{x-1}{x^2+2x}$ is a rational function, it is continuous at every point in its domain. The domain of $f(x)$ is the set $\left(-\infty, -2\right) \cup (-2, 0) \cup \left(0, +\infty\right)$. Thus, $f(x)$ is continuous over each of the intervals $\left(-\infty, -2\right)$, $(-2, 0)$, and $\left(0, +\infty\right)$.

EXAMPLE 2.34

Continuity over an Interval

State the interval(s) over which the function $f(x) = \sqrt{4 - x^2}$ is continuous.

Solution

From the limit laws, we know that $\lim\limits_{x \to a} \sqrt{4 - x^2} = \sqrt{4 - a^2}$ for all values of a in $(-2, 2)$. We also know that $\lim\limits_{x \to -2^+} \sqrt{4 - x^2} = 0$ exists and $\lim\limits_{x \to 2^-} \sqrt{4 - x^2} = 0$ exists. Therefore, $f(x)$ is continuous over the interval $[-2, 2]$.

✓ 2.24 State the interval(s) over which the function $f(x) = \sqrt{x + 3}$ is continuous.

The Composite Function Theorem allows us to expand our ability to compute limits. In particular, this theorem ultimately

allows us to demonstrate that trigonometric functions are continuous over their domains.

Theorem 2.9

Composite Function Theorem
If $f(x)$ is continuous at L and $\lim_{x \to a} g(x) = L$, then

$$\lim_{x \to a} f(g(x)) = f\left(\lim_{x \to a} g(x)\right) = f(L).$$

Before we move on to Example 2.35, recall that earlier, in the section on limit laws, we showed $\lim_{x \to 0} \cos x = 1 = \cos(0)$. Consequently, we know that $f(x) = \cos x$ is continuous at 0. In Example 2.35 we see how to combine this result with the composite function theorem.

EXAMPLE 2.35

Limit of a Composite Cosine Function
Evaluate $\lim_{x \to \pi/2} \cos\left(x - \frac{\pi}{2}\right)$.

Solution
The given function is a composite of $\cos x$ and $x - \frac{\pi}{2}$. Since $\lim_{x \to \pi/2} \left(x - \frac{\pi}{2}\right) = 0$ and $\cos x$ is continuous at 0, we may apply the composite function theorem. Thus,

$$\lim_{x \to \pi/2} \cos\left(x - \frac{\pi}{2}\right) = \cos\left(\lim_{x \to \pi/2} \left(x - \frac{\pi}{2}\right)\right) = \cos(0) = 1.$$

2.25 Evaluate $\lim_{x \to \pi} \sin(x - \pi)$.

The proof of the next theorem uses the composite function theorem as well as the continuity of $f(x) = \sin x$ and $g(x) = \cos x$ at the point 0 to show that trigonometric functions are continuous over their entire domains.

Theorem 2.10

Continuity of Trigonometric Functions
Trigonometric functions are continuous over their entire domains.

Proof

We begin by demonstrating that $\cos x$ is continuous at every real number. To do this, we must show that $\lim_{x \to a} \cos x = \cos a$ for all values of a.

$$\begin{aligned}
\lim_{x \to a} \cos x &= \lim_{x \to a} \cos((x - a) + a) && \text{rewrite } x = x - a + a \\
&= \lim_{x \to a} (\cos(x - a)\cos a - \sin(x - a)\sin a) && \text{apply the identity for the cosine of the sum of two angles} \\
&= \cos\left(\lim_{x \to a} (x - a)\right)\cos a - \sin\left(\lim_{x \to a} (x - a)\right)\sin a && \lim_{x \to a} (x - a) = 0, \text{ and } \sin x \text{ and } \cos x \text{ are continuous at } 0 \\
&= \cos(0)\cos a - \sin(0)\sin a && \text{evaluate } \cos(0) \text{ and } \sin(0) \text{ and simplify} \\
&= 1 \cdot \cos a - 0 \cdot \sin a = \cos a.
\end{aligned}$$

The proof that $\sin x$ is continuous at every real number is analogous. Because the remaining trigonometric functions may be expressed in terms of $\sin x$ and $\cos x$, their continuity follows from the quotient limit law.

□

As you can see, the composite function theorem is invaluable in demonstrating the continuity of trigonometric functions.

As we continue our study of calculus, we revisit this theorem many times.

The Intermediate Value Theorem

Functions that are continuous over intervals of the form $[a, b]$, where *a* and *b* are real numbers, exhibit many useful properties. Throughout our study of calculus, we will encounter many powerful theorems concerning such functions. The first of these theorems is the **Intermediate Value Theorem**.

Theorem 2.11

The Intermediate Value Theorem

Let *f* be continuous over a closed, bounded interval $[a, b]$. If *z* is any real number between $f(a)$ and $f(b)$, then there is a number *c* in $[a, b]$ satisfying $f(c) = z$ in Figure 2.38.

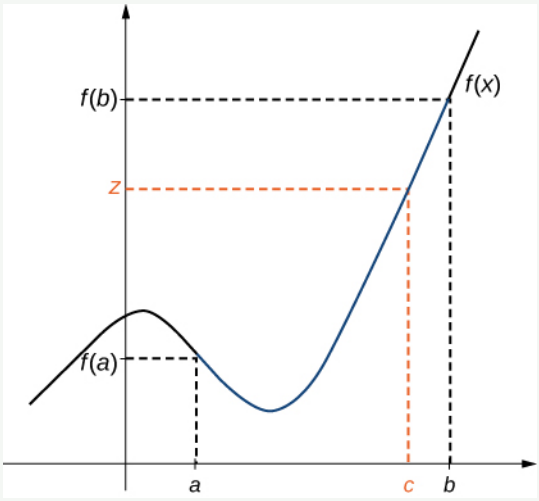


Figure 2.38 There is a number $c \in [a, b]$ that satisfies $f(c) = z$.

EXAMPLE 2.36

Application of the Intermediate Value Theorem

Show that $f(x) = x - \cos x$ has at least one zero.

Solution

Since $f(x) = x - \cos x$ is continuous over $\left(-\infty, +\infty\right)$, it is continuous over any closed interval of the form $[a, b]$. If you can find an interval $[a, b]$ such that $f(a)$ and $f(b)$ have opposite signs, you can use the Intermediate Value Theorem to conclude there must be a real number *c* in (a, b) that satisfies $f(c) = 0$. Note that

$$f(0) = 0 - \cos(0) = -1 < 0$$

and

$$f\left(\frac{\pi}{2}\right) = \frac{\pi}{2} - \cos\frac{\pi}{2} = \frac{\pi}{2} > 0.$$

Using the Intermediate Value Theorem, we can see that there must be a real number *c* in $[0, \pi/2]$ that satisfies $f(c) = 0$. Therefore, $f(x) = x - \cos x$ has at least one zero.

EXAMPLE 2.37

When Can You Apply the Intermediate Value Theorem?

If $f(x)$ is continuous over $[0, 2]$, $f(0) > 0$ and $f(2) > 0$, can we use the Intermediate Value Theorem to conclude that $f(x)$ has no zeros in the interval $[0, 2]$? Explain.

Solution

No. The Intermediate Value Theorem only allows us to conclude that we can find a value between $f(0)$ and $f(2)$; it doesn't allow us to conclude that we can't find other values. To see this more clearly, consider the function $f(x) = (x-1)^2$. It satisfies $f(0) = 1 > 0$, $f(2) = 1 > 0$, and $f(1) = 0$.

EXAMPLE 2.38

When Can You Apply the Intermediate Value Theorem?

For $f(x) = 1/x$, $f(-1) = -1 < 0$ and $f(1) = 1 > 0$. Can we conclude that $f(x)$ has a zero in the interval $[-1, 1]$?

Solution

No. The function is not continuous over $[-1, 1]$. The Intermediate Value Theorem does not apply here.

2.26 Show that $f(x) = x^3 - x^2 - 3x + 1$ has a zero over the interval $[0, 1]$.

SECTION 2.4 EXERCISES

For the following exercises, determine the point(s), if any, at which each function is discontinuous. Classify any discontinuity as jump, removable, infinite, or other.

131. $f(x) = \frac{1}{\sqrt{x}}$

132. $f(x) = \frac{2}{x^2+1}$

133. $f(x) = \frac{x}{x^2-x}$

134. $g(t) = t^{-1} + 1$

135. $f(x) = \frac{5}{e^x-2}$

136. $f(x) = \frac{|x-2|}{x-2}$

137. $H(x) = \tan 2x$

138. $f(t) = \frac{t+3}{t^2+5t+6}$

For the following exercises, decide if the function continuous at the given point. If it is discontinuous, what type of discontinuity is it?

139. $f(x) = \frac{2x^2-5x+3}{x-1}$ at $x = 1$

140. $h(\theta) = \frac{\sin\theta - \cos\theta}{\tan\theta}$ at $\theta = \pi$

141. $g(u) = \begin{cases} \frac{6u^2+u-2}{2u-1} & \text{if } u \neq \frac{1}{2} \\ \frac{7}{2} & \text{if } u = \frac{1}{2} \end{cases}$, at $u = \frac{1}{2}$

142. $f(y) = \frac{\sin(\pi y)}{\tan(\pi y)}$, at $y = 1$

143. $f(x) = \begin{cases} x^2 - e^x & \text{if } x < 0 \\ x - 1 & \text{if } x \geq 0 \end{cases}$, at $x = 0$

144. $f(x) = \begin{cases} x\sin(x) & \text{if } x \leq \pi \\ x\tan(x) & \text{if } x > \pi \end{cases}$, at $x = \pi$

In the following exercises, find the value(s) of k that makes each function continuous over the given interval.

145. $f(x) = \begin{cases} 3x + 2, & x < k \\ 2x - 3, & k \leq x \leq 8 \end{cases}$

146. $f(\theta) = \begin{cases} \sin\theta, & 0 \leq \theta < \frac{\pi}{2} \\ \cos(\theta + k), & \frac{\pi}{2} \leq \theta \leq \pi \end{cases}$

147. $f(x) = \begin{cases} \frac{x^2+3x+2}{x+2}, & x \neq -2 \\ k, & x = -2 \end{cases}$

148. $f(x) = \begin{cases} e^{kx}, & 0 \leq x < 4 \\ x + 3, & 4 \leq x \leq 8 \end{cases}$

149. $f(x) = \begin{cases} \sqrt{kx}, & 0 \leq x \leq 3 \\ x + 1, & 3 < x \leq 10 \end{cases}$

In the following exercises, use the Intermediate Value Theorem (IVT).

150. Let $h(x) = \begin{cases} 3x^2 - 4, & x \le 2 \\ 5 + 4x, & x > 2 \end{cases}$ Over the interval $[0, 4]$, there is no value of x such that $h(x) = 10$, although $h(0) < 10$ and $h(4) > 10$. Explain why this does not contradict the IVT.

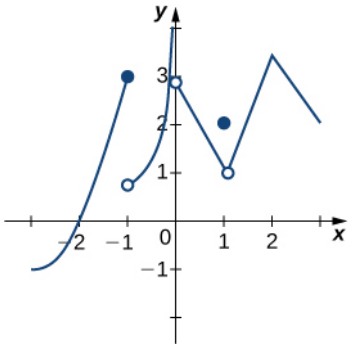
151. A particle moving along a line has at each time t a position function $s(t)$, which is continuous. Assume $s(2) = 5$ and $s(5) = 2$. Another particle moves such that its position is given by $h(t) = s(t) - t$. Explain why there must be a value c for $2 < c < 5$ such that $h(c) = 0$.

152. **[T]** Use the statement "The cosine of t is equal to t cubed."

a. Write a mathematical equation of the statement.
b. Prove that the equation in part a. has at least one real solution.
c. Use a calculator to find an interval of length 0.01 that contains a solution.

153. Apply the IVT to determine whether $2^x = x^3$ has a solution in one of the intervals $[1.25, 1.375]$ or $[1.375, 1.5]$. Briefly explain your response for each interval.

154. Consider the graph of the function $y = f(x)$ shown in the following graph.



a. Find all values for which the function is discontinuous.
b. For each value in part a., state why the formal definition of continuity does not apply.
c. Classify each discontinuity as either jump, removable, or infinite.

155. Let $f(x) = \begin{cases} 3x, x > 1 \\ x^3, x < 1 \end{cases}$.

a. Sketch the graph of f.
b. Is it possible to find a value k such that $f(1) = k$, which makes $f(x)$ continuous for all real numbers? Briefly explain.

156. Let $f(x) = \frac{x^4-1}{x^2-1}$ for $x \neq -1, 1$.
a. Sketch the graph of *f*.
b. Is it possible to find values k_1 and k_2 such that $f(-1) = k_1$ and $f(1) = k_2$, and that makes $f(x)$ continuous for all real numbers? Briefly explain.

157. Sketch the graph of a function $y = f(x)$ with properties i. through vi.
i. The domain of *f* is $\left(-\infty, +\infty\right)$.
ii. *f* has an infinite discontinuity at $x = -6$.
iii. $f(-6) = 3$
iv. $\lim_{x\to -3^-} f(x) = \lim_{x\to -3^+} f(x) = 2$
v. $f(-3) = 3$
vi. *f* is left continuous but not right continuous at $x = 3$.

158. Sketch the graph of a function $y = f(x)$ with properties i. through iv.
i. The domain of *f* is $[0, 5]$.
ii. $\lim_{x\to 1^+} f(x)$ and $\lim_{x\to 1^-} f(x)$ exist and are equal.
iii. $f(x)$ is left continuous but not continuous at $x = 2$, and right continuous but not continuous at $x = 3$.
iv. $f(x)$ has a removable discontinuity at $x = 1$, a jump discontinuity at $x = 2$, and the following limits hold: $\lim_{x\to 3^-} f(x) = -\infty$ and $\lim_{x\to 3^+} f(x) = 2$.

In the following exercises, suppose $y = f(x)$ is defined for all x. For each description, sketch a graph with the indicated property.

159. Discontinuous at $x = 1$ with $\lim_{x\to -1} f(x) = -1$ and $\lim_{x\to 2} f(x) = 4$

160. Discontinuous at $x = 2$ but continuous elsewhere with $\lim_{x\to 0} f(x) = \frac{1}{2}$

Determine whether each of the given statements is true. Justify your response with an explanation or counterexample.

161. $f(t) = \frac{2}{e^t - e^{-t}}$ is continuous everywhere.

162. If the left- and right-hand limits of $f(x)$ as $x \to a$ exist and are equal, then *f* cannot be discontinuous at $x = a$.

163. If a function is not continuous at a point, then it is not defined at that point.

164. According to the IVT, $\cos x - \sin x - x = 2$ has a solution over the interval $[-1, 1]$.

165. If $f(x)$ is continuous such that $f(a)$ and $f(b)$ have opposite signs, then $f(x) = 0$ has exactly one solution in $[a, b]$.

166. The function $f(x) = \frac{x^2-4x+3}{x^2-1}$ is continuous over the interval $[0, 3]$.

167. If $f(x)$ is continuous everywhere and $f(a), f(b) > 0$, then there is no root of $f(x)$ in the interval $[a, b]$.

[T] The following problems consider the scalar form of Coulomb's law, which describes the electrostatic force between two point charges, such as electrons. It is given by the equation $F(r) = k_e \frac{|q_1 q_2|}{r^2}$, where k_e is Coulomb's constant, q_i are the magnitudes of the charges of the two particles, and r is the distance between the two particles.

168. To simplify the calculation of a model with many interacting particles, after some threshold value $r = R$, we approximate *F* as zero.
   a. Explain the physical reasoning behind this assumption.
   b. What is the force equation?
   c. Evaluate the force *F* using both Coulomb's law and our approximation, assuming two protons with a charge magnitude of 1.6022×10^{-19} coulombs (C), and the Coulomb constant $k_e = 8.988 \times 10^9 \text{Nm}^2/\text{C}^2$ are 1 m apart. Also, assume $R < 1$ m. How much inaccuracy does our approximation generate? Is our approximation reasonable?
   d. Is there any finite value of *R* for which this system remains continuous at *R*?

169. Instead of making the force 0 at *R*, instead we let the force be 10^{-20} for $r \geq R$. Assume two protons, which have a magnitude of charge 1.6022×10^{-19} C, and the Coulomb constant $k_e = 8.988 \times 10^9 \text{Nm}^2/\text{C}^2$. Is there a value *R* that can make this system continuous? If so, find it.

Recall the discussion on spacecraft from the chapter opener. The following problems consider a rocket launch from Earth's surface. The force of gravity on the rocket is given by $F(d) = -mk/d^2$, where m is the mass of the rocket, d is the distance of the rocket from the center of Earth, and k is a constant.

170. **[T]** Determine the value and units of *k* given that the mass of the rocket is 3 million kg. (*Hint*: The distance from the center of Earth to its surface is 6378 km.)

171. **[T]** After a certain distance *D* has passed, the gravitational effect of Earth becomes quite negligible, so we can approximate the force function by

$$F(d) = \begin{cases} -\frac{mk}{d^2} & \text{if } d < D \\ 10{,}000 & \text{if } d \geq D \end{cases}.$$

Using the value of k found in the previous exercise, find the necessary condition *D* such that the force function remains continuous.

172. As the rocket travels away from Earth's surface, there is a distance *D* where the rocket sheds some of its mass, since it no longer needs the excess fuel storage. We can write this function as

$$F(d) = \begin{cases} -\frac{m_1 k}{d^2} \text{ if } d < D \\ -\frac{m_2 k}{d^2} \text{ if } d \geq D \end{cases}.$$

Is there a *D* value such that this function is continuous, assuming $m_1 \neq m_2$?

Prove the following functions are continuous everywhere

173. $f(\theta) = \sin\theta$

174. $g(x) = |x|$

175. Where is

$$f(x) = \begin{cases} 0 \text{ if } x \text{ is irrational} \\ 1 \text{ if } x \text{ is rational} \end{cases}$$

continuous?